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Beyond Repeated Sampling

Learning Search Policies for LLM Reasoning via Reinforcement-Trained Concept Generation

By Ismail Labiad, Matthieu Kowalski, Marc Schoenauer, Rémi Munos & Julia Kempe (Meta FAIR)

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A small concept generator, trained with reinforcement learning to maximize the pass@k of a frozen large answer generator, achieves substantially higher diversity and downstream accuracy than naive repeated sampling at the same compute budget—and transfers to answer generators it was never trained against, including those from a different model family.

AT A GLANCE

Problem: Naive repeated sampling for test-time compute explores only through local decoding noise, producing near-duplicate attempts rather than semantically diverse solutions.

Why it matters: Hard reasoning problems require qualitatively different strategies; pass@k saturates quickly when all samples share the same implicit approach.

Method: Train a small concept generator with RL to emit problem-specific hints and strategies that maximize downstream success of a frozen large answer generator.

Result: Trained concept generator substantially outperforms repeated sampling and untuned large models; transfers to unseen answer generators.

Evidence: Hard mathematical reasoning benchmarks at equal answer-generation compute budget.

The Big Picture

The dominant paradigm for allocating test-time compute to large language models is naive repeated sampling: draw k independent answers from the model at high temperature, then select the best by a verifier or majority vote. This strategy is seductively simple and, at moderate k , effective. But it has a fundamental limitation: each sample explores only through local decoding noise. Temperature and top-p diversify token choices within a trajectory, but they cannot steer the model toward qualitatively different approaches to the same problem.

On hard mathematical reasoning tasks—where correct

solutions often require an insight, decomposition, or analogy that is far from the model’s prior mode—repeated sampling saturates quickly. The k -th sample is nearly as redundant as the second: the model keeps attempting the same dominant strategy with minor surface variations.

This paper asks whether test-time exploration can instead be steered at a *semantic* level. The answer is yes, and the key is to separate the role of *exploration* (finding diverse approaches) from *execution* (implementing a given approach): a small, trainable concept generator proposes strategies; a large, frozen answer generator executes them.

Why Existing Approaches Fall Short

Repeated sampling (best-of- N , majority vote) explores only through decoding stochasticity. Diversity is local; semantically distinct strategies require explicit steering.

Self-consistency aggregates repeated samples but does not improve the diversity of those samples—it is a better aggregator, not a better explorer.

Tree search (MCTS) explores explicitly but requires a well-trained per-step verifier, which is expensive to obtain and domain-specific.

Prompting for diversity (e.g., asking the model to use different methods) provides one alternative per call; emitting a library of diverse concepts in a single trajectory is more efficient.

Scaling the answer generator at inference time is expensive; the goal is to make a fixed frozen model explore more effectively with a lightweight companion.

Core Mechanism

Concept-Conditioned Generation. Given a problem x , the method first samples a *concept* c —a natural-language hint, strategy keyword, or approach description—then generates an answer from the large model conditioned on (x, c) . The concept shifts the answer generator toward a different region of solution space.

Multi-Concept Trajectories. Rather than one concept per call, the concept generator emits a diverse list of concepts in a single forward pass, covering different strategies before any answer is generated. This maximizes spread at minimal additional cost.

Reinforcement-Learned Concept Generator. A small model π_θ is trained with RL to emit concepts that maximize the downstream binary reward (answer correctness). The reward flows back to the concept generator but not to the frozen answer generator f .

Technical Formulation

Let x denote a problem, c a concept sampled from $\pi_\theta(\cdot | x)$, and f the frozen large answer generator. The concept-conditioned pass@k:

$$\text{pass@k}(x) = 1 - \frac{\binom{n-s}{k}}{\binom{n}{k}}$$

where n total samples are generated and s are correct. The RL objective:

$$\max_{\theta} \mathbb{E}_{x \sim \mathcal{D}} [\text{pass@k}_{c \sim \pi_\theta(\cdot | x)}(f(x, \cdot))]$$

The policy gradient update uses binary correctness as the scalar reward:

$$\nabla_{\theta} \mathcal{L}(\theta) = -\mathbb{E}_{c \sim \pi_{\theta}} [R(x, c) \nabla_{\theta} \log \pi_{\theta}(c | x)]$$

where $R(x, c) = \mathbf{1}[f(x, c) \text{ is correct}]$ (or a normalized group-relative variant for variance reduction).

Learning or Inference Procedure

Training:

1. Initialize π_{θ} from a small pretrained LM.
2. For each training problem x : sample K concepts from π_{θ} , query f for each, record correctness.
3. Update π_{θ} with policy gradient on the correctness reward.

Inference:

1. Sample a batch of concepts $c_1, \dots, c_K \sim \pi_{\theta}(\cdot | x)$.
2. Generate answers $a_i = f(x, c_i)$ for each concept.
3. Return the most frequent or highest-confidence answer.

What the Guarantee Says

The trained concept generator is a reusable search policy: it does not learn to solve problems directly but learns which types of strategies lead the answer generator to success on the training distribution. Transfer to new answer generators follows because the learned strategies are problem-structural rather than model-specific.

Experimental Findings

- Hard mathematical reasoning benchmarks where repeated sampling pass@k plateaus at high k
- Concept-conditioned generation with trained π_{θ} achieves substantially higher pass@k than naive repeated sampling at equal answer-generation compute budget
- Trained small generator outperforms concepts from untuned models many times larger, confirming that RL training—not model scale—is the key factor

- Zero-shot transfer to an answer generator from a different model family succeeds: concepts learned for one model improve another
- Multi-concept trajectories (many concepts per call) further improve pass@k by reducing concept redundancy

Ablations and Interpretation

Single vs. multi-concept: Emitting a diverse list of concepts in one trajectory reduces redundancy and improves pass@k compared to sampling one concept per call.

Generator size: Smaller trained generators outperform larger zero-shot generators, demonstrating the value of training over scale.

Reward signal: Binary correctness alone is sufficient as the RL reward; no dense process reward model or verifier is needed during training.

Transfer: Concept quality transfers across answer generators from different model families, suggesting that the search policy captures problem-structural strategies rather than artifacts of a specific model’s output distribution.

Reference

Ismail Labiad, Matthieu Kowalski, Marc Schoenauer, Rémi Munos, Julia Kempe. **Beyond Repeated Sampling: Learning Search Policies for LLM Reasoning.** arXiv:2609.26704, September 2026.

<https://arxiv.org/abs/2609.26704>

Keywords: mechanistic interpretability • hallucination • sparse probing • neuron localization

Hallucination Neurons and Where to Find Them

An Investigation into the Existence of Hallucination Neurons in LLMs via Diagnostic Protocol

By (authors not specified in pre-print metadata)

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A five-step diagnostic protocol reveals that 19 of 22 claimed hallucination neurons in Gemma 3 4B have Pearson $|r| > 0.7$ with other features, exhibit only moderate bootstrap stability, and show weak sparse-dense ranking overlap — demonstrating that sparse predictive structure coexists with non-unique neuron selection, and that routine diagnostic validation is necessary before making causal localization claims.

AT A GLANCE

Problem: Sparse probing claims specific neurons causally drive LLM hallucination, but these claims are not tested against known failure modes of L1-regularized probing in correlated feature spaces.

Why it matters: Non-unique neuron selection undermines the validity of targeted steering and auditing based on claimed hallucination neurons.

Method: A five-step diagnostic protocol: feature correlation, bootstrap stability, sparse vs. dense ranking, intervention baselines, and cross-dataset evaluation.

Result: 19/22 H-Neurons fail the feature correlation criterion in Gemma 3 4B; bootstrap stability and sparse-dense agreement are weak.

Evidence: TriviaQA, BioASQ, and NQ-Open across Gemma 3 4B and MedGemma 4B.

The Big Picture

Mechanistic interpretability of large language models has made rapid progress in identifying circuits, features, and neurons associated with specific behaviors. A particularly consequential line of work claims to identify “hallucination neurons”—small sets of neurons whose activation levels predict and causally drive factual hallucinations. If such neurons exist and are uniquely identifiable, they offer a compelling target for behavioral steering: suppress the neuron, suppress the hallucination.

But neuron identification typically relies on L1-regularized (Lasso-style) sparse probing, which is well-known to select non-unique solutions in the presence of correlated features. A Lasso probe that achieves high AUROC does not estab-

lish that the selected neurons are the unique causal locus of the behavior—only that they are predictive given the regularization and the training data. In high-dimensional, highly correlated neural activation spaces, this distinction matters enormously.

This paper proposes a minimum diagnostic standard for neuron-localization claims and applies it to the H-Neuron literature, revealing systematic non-uniqueness in the claimed hallucination neurons.

Why Existing Approaches Fall Short

L1-regularized probing selects solutions non-uniquely when features are correlated: many alternative neuron sets achieve similar or better discrimination. The selected set depends on regularization strength, initialization, and the specific training instances.

High AUROC is a necessary but not sufficient condition for localization. Many overlapping neuron subsets may achieve high AUROC; the probe’s predictive success does not establish which neurons are causally necessary.

Ablation studies (removing a selected neuron and measuring behavior change) are often compared against no-ablation baselines rather than against equally predictive alternative neurons, making causal claims fragile.

Single-dataset evaluation cannot distinguish neurons that detect hallucination generically from neurons whose predictive power is specific to the evaluation dataset’s format or domain.

Core Mechanism

Five-Step Diagnostic Protocol. The paper proposes five diagnostic criteria that should be satisfied before claiming unique neuron localization:

1. **Feature correlation:** Flag selected neurons with Pearson $|r| > 0.7$ with unselected features as potentially non-unique.
2. **Bootstrap stability:** Measure Jaccard overlap of the selected neuron set across bootstrap samples of the probe training data.
3. **Sparse vs. dense ranking:** Compare the ranking of neurons by the L1 probe with their ranking by a dense (non-regularized) importance score; low overlap signals L1-driven selection, not importance.
4. **Intervention baselines:** Compare ablation of selected neurons against ablation of equally predictive randomly drawn neurons; a uniquely causal neuron should show strictly greater behavior change.
5. **Cross-dataset evaluation:** Evaluate selected neurons on held-out datasets not used for selection; poor transfer indicates dataset-specific rather than general hallucination neurons.

Technical Formulation

Let $\mathbf{h} \in \mathbb{R}^d$ be activations at a layer, $y \in \{0, 1\}$ the hallucination label. The L1-regularized probe solves:

$$\hat{\beta} = \arg \min_{\beta} \mathcal{L}(y, \mathbf{h}^\top \beta) + \lambda \|\beta\|_1$$

Selected neuron set: $S = \{i : \hat{\beta}_i \neq 0\}$.

Criterion 1 (feature correlation): Flag S if $\exists j \notin S, i \in S$ such that $|\text{Corr}(h_i, h_j)| > 0.7$.

Criterion 2 (bootstrap stability): $J_B = |S \cap S_b| / |S \cup S_b|$ for bootstrap sample b ; flag if $\mathbb{E}[J_B] < 0.8$.

Criterion 3 (ranking disagreement): Spearman $\rho(\hat{\beta}, \tilde{\beta})$ where $\tilde{\beta}$ is the dense-probe importance; flag if $\rho < 0.5$.

Cross-dataset AUROC gaps (Gemma 3 4B vs MedGemma 4B):

$$\Delta_{\text{TriviaQA}} = +0.311 \text{ vs } +0.235$$

$$\Delta_{\text{BioASQ}} = +0.474 \text{ vs } +0.455$$

$$\Delta_{\text{NQ-Open}} = +0.128 \text{ vs } +0.112$$

Learning or Inference Procedure

The diagnostic is applied post-hoc to trained probes:

1. Train an L1 probe on each (model, dataset) pair.
2. Record the selected neuron set S .
3. Apply each diagnostic criterion.
4. Report pass/fail and overall diagnostic verdict.

No model training is performed; the protocol is a validation framework.

What the Guarantee Says

When all five criteria are satisfied, the selected neuron set is consistent with being a unique causal locus of the behavior. When criteria 1–3 fail (as they do here), the probe’s predictive power is consistent with a broader class of alternative neuron sets, and the causal localization claim is not established. This does not falsify hallucination detection; it shows the detection is distributed, not localized.

Experimental Findings

- **19 of 22** H-Neurons in Gemma 3 4B have Pearson $|r| > 0.7$ with at least one unselected feature—failing criterion 1
- Bootstrap Jaccard overlap is moderate, confirming sensitivity to training subsample—failing criterion 2
- Sparse-dense rank correlation is weak, indicating L1 regularization drives the selection—failing criterion 3

- Gemma 3 4B consistently outperforms MedGemma 4B on hallucination probe AUROC across all three datasets
- Cross-dataset probe quality varies, suggesting partial dataset specificity in neuron selection

Ablations and Interpretation

AUROC vs. uniqueness: High AUROC is achieved by many overlapping neuron sets; the reported probe AUROC does not support localization.

Dataset specificity: Different datasets select partially different neuron sets, consistent with distributed rather than universal hallucination neurons.

Implications for steering: Interventions on L1-selected neurons may or may not generalize because many correlated alternatives exist. Robust steering likely requires denser probes, ensemble methods, or causal identification beyond predictive accuracy.

Broader message: Routine diagnostic validation—particularly feature correlation and bootstrap stability checks—should be a minimum requirement for sparse-neuron localization claims in interpretability research.

Reference

Hallucination Neurons and Where to Find Them: An Investigation into the Existence of Hallucination Neurons. arXiv:2609.29781, September 2026.
<https://arxiv.org/abs/2609.29781>

Keywords: scaling laws • inference-time compute • chain-of-thought • Bayesian analysis

Scaling of Capability and Efficiency

Joint Quantitative Laws for Solve Rate and Token Cost in Large Reasoning Models

By (authors not specified in pre-print metadata)

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Using hierarchical Bayesian models applied to the DeepSeek-R1-Distill family, this paper shows that solve probability decays exponentially with problem hardness while the characteristic scale grows as a power law with model parameters—and that among solved instances, output length grows as a power law with instance size, providing the first joint quantitative account of capability and efficiency scaling at inference time.

AT A GLANCE

Problem: Capability and efficiency in chain-of-thought reasoning both depend on problem hardness and model size, but their joint scaling behaviour has not been quantitatively characterized.

Why it matters: Without joint scaling laws, predicting compute requirements for a given problem difficulty and model size is guesswork.

Method: Hierarchical Bayesian models fit to the DeepSeek-R1-Distill family across four algorithmic problem classes with controlled hardness.

Result: Exponential capability decay with power-law characteristic scale; power-law token cost growth among correctly solved instances.

Evidence: 1.5B–70B parameter DeepSeek-R1-Distill models on arithmetic, multiplication, sorting, and path-finding.

The Big Picture

Large reasoning models (LRMs) achieve state-of-the-art results by generating extended chain-of-thought traces before producing final answers. The amount of computation consumed at inference time—measured in output tokens—grows substantially with problem difficulty, and harder problems are solved by larger models. Both capability (solve rate) and efficiency (tokens per correct solution) thus depend on at least two variables: problem hardness and model size.

Despite the obvious importance of understanding this dependency, it has not been characterized in the form of

quantitative scaling laws. Practitioners must therefore guess how many tokens a given model will require to solve problems of a given difficulty, and how much larger a model must be to solve problems that the current model cannot.

This paper provides the first joint quantitative account of capability and efficiency scaling at inference time, using controlled algorithmic problem classes where hardness is operationalized as instance size and correctness is verified exactly.

Why Existing Approaches Fall Short

Average accuracy evaluations aggregate performance across problem difficulties, masking the exponential decay with hardness and conflating easy-instance dominance with true capability on hard instances.

Single-model comparisons at fixed compute budgets cannot reveal how model scale shifts the capability-efficiency frontier.

Simple regression on token counts ignores the joint structure of success and token cost, treating failed attempts identically to efficient successes.

Prior scaling laws (Chinchilla, etc.) address training compute and not inference-time token generation conditioned on problem difficulty, leaving the test-time compute dimension uncharacterized.

Core Mechanism

Hierarchical Bayesian Model. The paper fits a generative model to solve indicators and token counts across models and problem instances. The model has three layers:

1. **Instance level:** binary solve indicator and token count per instance per model.
2. **Class level:** partial pooling of scaling exponents across four problem classes.
3. **Model level:** power-law relationship between model size and scaling parameters.

Problem Classes. Four algorithmic tasks with exact verifiers: arithmetic computation, digit multiplication, list sorting, and path-finding. Hardness is controlled by instance size n (digits, list length, or graph nodes).

Model Family. DeepSeek-R1-Distill at 1.5B, 7B, 14B, 32B, and 70B parameters—a controlled family where differences arise from model scale alone.

Technical Formulation

Solve probability at instance size n and model parameter count m :

$$p(n, m) = p_0(m) \cdot \exp\left(-\frac{n}{\lambda(m)}\right)$$

Characteristic scale as a power law in model parameters:

$$\lambda(m) = \alpha \cdot m^\beta$$

Conditional output length (correctly solved instances):

$$\mathbb{E}[L \mid \text{solved}, n, m] = \gamma(m) \cdot n^\delta$$

Hierarchical priors (partial pooling across problem classes $k = 1, \dots, K$):

$$\beta_k \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2), \quad \delta_k \sim \mathcal{N}(\mu_\delta, \sigma_\delta^2)$$

All parameters are jointly estimated by MCMC, yielding posterior distributions over scaling exponents β and δ per class.

Learning or Inference Procedure

1. For each (model, problem class, instance size): generate up to R attempts; record solve indicator and output token count.
2. Fit the hierarchical Bayesian model to the full dataset using MCMC (Stan or equivalent).
3. Report posterior means and 90% credible intervals for $\alpha, \beta, \gamma, \delta$ per class.
4. Use fitted model to predict solve probability and expected token cost for new (model size, instance size) pairs.

What the Guarantee Says

The hierarchical model provides posterior uncertainty over scaling exponents. Within the linear-arithmetic and combinatorial-algorithmic problem classes studied, exponential solve-rate decay with power-law characteristic scale is a consistent structural finding, not an artifact of a single model or problem class. Cross-class partial pooling provides regularization that stabilizes estimates at large difficulty where data is sparse.

Experimental Findings

- **5 models:** 1.5B to 70B DeepSeek-R1-Distill
- **4 problem classes:** arithmetic, digit multiplication, list sorting, path-finding
- Solve probability fits exponential decay with hardness ($R^2 > 0.92$) across all classes

- Characteristic scale exponent $\beta \approx 0.3\text{--}0.5$ by class; path-finding has the steepest decay
- DeepSeek-R1-Distill-70B achieves approximately $10\times$ larger characteristic scale than the 1.5B model on arithmetic
- Output length exponent $\delta \approx 1.0\text{--}1.5$: near-linear to super-linear growth with instance size
- Efficiency degrades faster than capability with hardness: diminishing returns to scale on the hardest instances

Ablations and Interpretation

Problem class differences: Path-finding shows the steepest capability decay, suggesting it requires qualitatively more extensive search. Arithmetic shows the shallowest, consistent with near-algorithmic solutions.

Pooling vs. no pooling: Hierarchical pooling across classes stabilizes estimates at large difficulty; without pooling, estimates at the hardest difficulty levels have large variance.

Efficiency vs. capability trade-off: The power-law exponent $\delta > 1$ for output length implies that even when a larger model can solve harder instances, it pays a super-linear token cost. Compute-optimal scaling must balance capability and efficiency gains.

Implications: Given model size m and a target solve probability p^* on instances of size n^* , the minimum required model size is: $m^* = (\lambda^{-1}(-n^*/\ln p^*)/\alpha)^{1/\beta}$ —a prediction the framework makes interpretable and calibrated.

Reference

Scaling of Capability and Efficiency at Inference Time in Large Reasoning Models. arXiv:2609.27166, September 2026.

<https://arxiv.org/abs/2609.27166>

Keywords: robot manipulation • LLM planning • adaptive control • physical state

Body-Grounded Replanning

Physically Adaptive Robot Manipulation via LLM-Based Internal State Replanning

By Namiko Saito & Hiroshi Kera (Microsoft Research Asia Tokyo; Waseda University; Chiba University; National Institute of Informatics)

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Body-grounded high-level replanning uses internal physical signals—joint torques and mobility constraints—to trigger LLM-based strategy switching during manipulation, maintaining task success while substantially reducing physical effort under load and asymmetric mobility conditions.

AT A GLANCE

Problem: A manipulation strategy may remain geometrically feasible while becoming physically unsuitable due to increased joint load or limited mobility; current planners have no mechanism to detect or respond to this.

Why it matters: Robots deployed in real environments accumulate physical state changes that degrade performance and increase wear; without adaptation, task success and hardware longevity both suffer.

Method: Body-state events trigger an LLM that interprets joint-level state and execution history to select a physically appropriate alternative strategy from a library, without changing the task objective or low-level controller.

Result: High task success maintained; physical effort reduced; more efficient strategy adaptation than without body state awareness.

Evidence: Reaching task under controlled load and asymmetric mobility constraints, simulation and real robot.

The Big Picture

Robotic manipulation planners reason carefully about the external world: obstacle geometry, object poses, grasp feasibility, and collision avoidance. But they typically treat the robot’s own body as a fixed, ideal actuator whose physical state does not change during execution.

In practice, this assumption fails. A joint under sustained load accumulates fatigue. A component subject to hardware wear constrains the workspace asymmetrically. A strategy that was geometrically and dynamically

feasible at planning time may become physically unsuitable minutes later. The standard response is to run the low-level controller harder—increasing torque outputs to compensate—but this accelerates wear, increases energy consumption, and can ultimately cause failure.

Body-grounded replanning introduces a new level in the robot’s deliberative hierarchy: the ability to recognize when internal physical state has made the current high-level strategy inappropriate, and to select a physically better alternative without changing the task objective or the low-level controller.

Why Existing Approaches Fall Short

Standard task-and-motion planning (TAMP) optimizes over geometry and kinematics. Joint torque, fatigue, and mobility constraints are not modeled at the planning level; they are delegated to the controller.

Adaptive controllers adjust joint-level torques and impedances in response to physical state but do not change high-level strategy (e.g., which direction to approach, which grasp type to use).

LLM-based planners condition on task descriptions and environmental observations—scene graphs, object attributes, occupancy maps—but have no access to joint-level physical signals.

Fault-tolerant planning handles discrete, identifiable hardware failures (a joint locked at zero torque) but not continuous, graded physical state degradation.

Core Mechanism

Body-State Event Detection. Physical state $s_t = (\tau_t, q_t, v_t)$ is monitored continuously. An event triggers when:

- Joint load: $\max_j \tau_{j,t} > \tau_{\text{thresh}}$
- Mobility constraint: reachability $R(q_t) < R_{\text{thresh}}$

LLM Strategy Selector. On event, the LLM receives a prompt encoding:

1. **Joint-level state:** current torques and a natural-language summary of load distribution and mobility limits.
2. **Execution statistics:** success rate and mean completion time of recent strategy attempts.
3. **Execution history:** which strategies were tried and with what outcomes.

The LLM selects a new strategy from the strategy library (e.g., “approach from the side,” “use extended-arm reach,” “shift load to stronger joints”).

Decoupled Architecture. High-level replanning and low-level execution are fully decoupled: replanning changes the strategy but not the task objective or the

controller’s internal parameters.

Technical Formulation

Let $\sigma \in \mathcal{S}$ be the current strategy from library $\mathcal{S}$. Event condition at time t :

$$e_t = \mathbf{1} \left[\max_j \tau_{j,t} > \tau_{\text{thresh}} \vee R(q_t) < R_{\text{thresh}} \right]$$

On $e_t = 1$, the LLM is queried:

$$\sigma_{t+1} = \text{LLM}(\mathcal{P}(s_{t-T:t}, h_{0:t}, \sigma_{\text{hist}}))$$

where $\mathcal{P}$ encodes physical state over window T , execution history h , and prior strategies σ_{hist} .

The optimization objective:

$$\max_{\sigma_{0:\infty}} \Pr[\text{task success}] \text{ s.t. } \sum_t \|\tau_t\|_2 \leq E_{\text{budget}}$$

Learning or Inference Procedure

No model training is required. At execution time:

1. Execute strategy σ with the low-level controller.
2. Monitor s_t at each control step.
3. On $e_t = 1$: construct prompt $\mathcal{P}$, query the LLM.
4. Update $\sigma \leftarrow \sigma_{t+1}$ and resume.
5. Repeat until success or timeout.

The LLM runs at inference time only, with no task-specific fine-tuning.

What the Guarantee Says

The framework does not require training a new model. Its correctness relies on: (i) the strategy library covering alternative strategies that are physically better in the relevant conditions, and (ii) the LLM’s ability to interpret physical state descriptions and select appropriate alternatives. Evaluation shows that both conditions hold for the tested configurations.

Experimental Findings

- **Task:** Reaching under two constraint types—controlled joint load (added deadweight) and asymmetric mobility (reduced joint range)
- **Platforms:** Simulated and real robot arm
- Body-grounded replanning maintains high task success rates where fixed-strategy baselines fail under increasing load
- Cumulative joint torque is substantially reduced by selecting strategies that distribute load across available joints

- Strategy adaptation is more efficient (fewer failed attempts before success) than random strategy switching when mobility is asymmetric
- Performance is robust across multiple off-the-shelf LLMs; stronger models provide modestly better strategy selection

Ablations and Interpretation

Event threshold: Too low τ_{thresh} causes excessive re-planning; too high causes physical overload. Performance is robust within a reasonable range.

History window: More execution history improves strategy selection for mobility constraints; load constraints show less sensitivity.

Strategy library size: A larger library with more alternatives improves success under diverse constraints; the LLM selects appropriately among them.

Architecture decoupling: The clean separation between replanning and execution means the framework can be added to any existing manipulation system without modifying the low-level controller.

Reference

Namiko Saito, Hiroshi Kera. **Body-Grounded Replanning for Physically Adaptive Manipulation**. arXiv:2609.30024, September 2026.
<https://arxiv.org/abs/2609.30024>

Keywords: diffusion models • flow matching • latent space • representation learning

On the Diffusibility of High-Dimensional Latents

Reconstruction-Finetuned Encoders Collapse to Low-Dimensional Manifolds—Switch to Data Prediction to Fix Flow Matching

By Chao Feng, Zhiyang Xu, Bowei Chen, Yuanjun Xiong, Xiyao Wang, Jui-Hsien Wang, Richard Zhang, Zhe Lin, Andrew Owens & Yijun Li

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Reconstruction-finetuned visual encoders exhibit a sharp collapse in effective dimensionality, leaving standard velocity prediction in flow matching to fit orthogonal noise directions outside the signal manifold; switching to $\mathbf{x}_0$ -prediction (clean data parameterization) focuses optimization on the true signal subspace and substantially improves generation quality in high-dimensional Representation Autoencoder latent spaces.

AT A GLANCE

Problem: Finetuning visual encoders for reconstruction collapses their effective dimensionality, making standard velocity prediction in flow matching inefficient because it must fit noise directions orthogonal to the signal manifold.

Why it matters: Using pretrained visual encoder feature spaces as diffusion latents is desirable for rich semantic representations, but generation quality is poor without addressing this geometric mismatch.

Method: Analyze effective dimensionality collapse in reconstruction-finetuned RAEs; show $\mathbf{x}_0$ -prediction avoids fitting orthogonal noise directions.

Result: Substantial improvement in FID and CLIP scores by switching parameterization without any other change.

Evidence: CLIP ViT-L/14 and DINOv2 RAEs, ECCV 2026 evaluation.

The Big Picture

Latent diffusion models (LDMs) achieve high-quality image generation by operating in the compressed latent space of a variational autoencoder (VAE) rather than pixel space. A natural extension is to use the feature spaces of powerful pretrained visual encoders—CLIP, DINOv2, and their variants—as the latent space, combining the generative power of diffusion with the rich semantic representations learned by these encoders.

Representation Autoencoders (RAEs) equip pretrained encoders with a learned decoder to enable image reconstruction from features. But many off-the-shelf encoders are trained for classification or contrastive objectives and discard fine-grained visual details. To enable high-quality reconstruction, RAEs fine-tune the encoder on reconstruction objectives.

This finetuning introduces a subtle but consequential geometric problem: while the nominal feature dimension remains large (768, 1024, or more), the learned features concentrate near a much lower-dimensional subspace. Standard flow matching with velocity prediction must fit noise directions orthogonal to this subspace, making optimization inefficient and generation quality poor. This paper diagnoses the problem and proposes a simple, principled fix.

Why Existing Approaches Fall Short

Standard VAE latents (Stable Diffusion) are designed for diffusion: their dimensionality is explicitly controlled and their geometry is well-conditioned. RAE latents inherit neither property by design.

Velocity prediction (v -prediction, ε -prediction) in flow matching assumes data fills the ambient space. When data concentrates on a low-dimensional manifold, the velocity field has large orthogonal components that carry no signal but must be learned, wasting capacity.

Higher nominal dimensionality makes this worse: more dimensions implies more orthogonal directions to fit, even though no additional signal is present.

Dimensionality reduction (PCA projection before diffusion) partially addresses the problem but requires separate engineering, discards representation structure, and introduces artifacts at reconstruction.

Core Mechanism

Effective Dimensionality Analysis. Feature dimensionality is quantified via the participation ratio of the feature covariance:

$$d_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}$$

where λ_i are the eigenvalues of the feature covariance matrix. RAE features show a sharp collapse: $d_{\text{eff}} \ll d_{\text{nominal}}$ —typically 5–15% of the nominal dimension after reconstruction finetuning.

$\mathbf{x}_0$ -Prediction. In flow matching with linear interpolation:

$$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

Velocity target: $\mathbf{v} = \mathbf{x}_0 - \boldsymbol{\varepsilon}$, which has a large noise component $-\boldsymbol{\varepsilon}$ that spans the full d -dimensional space,

forcing the model to fit orthogonal directions.

Data target: $\mathbf{x}_0$, which lies on the d_{eff} -dimensional signal manifold. Predicting $\mathbf{x}_0$ avoids the orthogonal noise component entirely.

Technical Formulation

Let $\mathbf{x}_0 \in \mathbb{R}^d$ be an RAE feature with $d_{\text{eff}} \ll d$, and $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.

The noisy sample at time t :

$$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \varepsilon$$

Velocity-prediction loss:

$$\mathcal{L}_v = \mathbb{E}_{t, \mathbf{x}_0, \varepsilon} [\|f_\theta(\mathbf{x}_t, t) - (\mathbf{x}_0 - \varepsilon)\|^2]$$

Data-prediction loss:

$$\mathcal{L}_{x_0} = \mathbb{E}_{t, \mathbf{x}_0, \varepsilon} [\|f_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2]$$

The velocity residual decomposes as:

$$\mathbf{x}_0 - \varepsilon = \underbrace{\mathbf{x}_0}_{\text{signal, on manifold}} - \underbrace{\varepsilon}_{\text{noise, spans } \mathbb{R}^d}$$

Fitting $\mathbf{x}_0$ only is equivalent to solving a projection onto the signal manifold, avoiding the orthogonal noise component. The optimization landscape for $\mathcal{L}_{x_0}$ is lower-dimensional and better-conditioned than for $\mathcal{L}_v$ when $d_{\text{eff}} \ll d$.

Learning or Inference Procedure

Training:

1. Finetune the pretrained visual encoder as a Representation Autoencoder (RAE) for reconstruction quality.
2. Extract RAE features for the training image set.
3. Compute d_{eff} to characterize the signal manifold.
4. Train a flow-matching generator with $\mathcal{L}_{x_0}$ instead of $\mathcal{L}_v$.

Inference:

1. Sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.
2. Integrate the learned flow from $t = T$ to $t = 0$ using the $\mathbf{x}_0$ prediction.
3. Decode the generated feature $\mathbf{x}_0$ with the RAE decoder.

What the Guarantee Says

When $d_{\text{eff}} \ll d$, $\mathcal{L}_v$ is provably harder to optimize than $\mathcal{L}_{x_0}$ because the orthogonal noise components constitute a high-dimensional regression problem with no signal—they

can only be fitted by memorization. The data parameterization eliminates this by construction. The benefit scales with d/d_{eff} : the greater the effective dimensionality collapse, the larger the expected improvement.

Experimental Findings

- Evaluated with CLIP ViT-L/14 and DINOv2 RAEs at feature dimensions 768 and 1024
- d_{eff} is 5–15% of d_{nominal} after reconstruction finetuning
- $\mathbf{x}_0$ -prediction achieves substantially better FID compared to velocity prediction in the same RAE latent space
- CLIP similarity scores also improve, confirming benefit beyond distributional statistics
- Results are robust across encoder types and RAE decoder architectures
- FID gap between parameterizations correlates with d/d_{eff} , directly supporting the manifold mismatch hypothesis

Ablations and Interpretation

No RAE finetuning (frozen encoder): When the encoder is not finetuned for reconstruction, d_{eff} is higher and the gap between parameterizations is smaller, confirming the mechanism.

Alternative fix (PCA projection): Projecting to the principal subspace before velocity prediction partially recovers quality but is less elegant: it requires separate PCA fitting, discards features, and underperforms $\mathbf{x}_0$ -prediction.

Time t dependence: The difficulty of velocity prediction is greatest at intermediate t values where ε and $\mathbf{x}_0$ contribute equally; $\mathbf{x}_0$ -prediction remains well-conditioned throughout.

Nominal vs. effective dimensionality: Higher nominal dimension is strictly worse under velocity prediction but not under $\mathbf{x}_0$ -prediction, removing the disincentive to use richer representations.

Reference

Chao Feng, Zhiyang Xu, Bowei Chen, Yuanjun Xiong, Xiyao Wang, Jui-Hsien Wang, Richard Zhang, Zhe Lin, Andrew Owens, Yijun Li. **On the Diffusibility of High-Dimensional Latents**. arXiv:2609.28473, ECCV 2026. <https://arxiv.org/abs/2609.28473>